NASA SPACE APPS CAIRO 2026 · AI & MACHINE LEARNING

HOW TO WIN SPACE APPS WITH AI

From NASA data to an honest model, in 48 hours

AbdElRhman ElMoghazy · Senior Data Scientist

ABOUT ME

AbdElRhman ElMoghazy

Senior Data Scientist

Senior Data Scientist at Fertiglobe and an AI educator with 5+ years of experience in machine learning and generative AI.

Master's degree from Queen's University

Mentored 100+ learners, and delivered 80+ technical sessions.

Participated in multiple competitions and won first place in the MEA region at IBM's Call for Code 2019, sharing his passion for practical AI learning and innovation.

YOU LEAVE WITH

1. A **scope card** for your own idea
2. A **working notebook** you ran yourself
3. One **honest claim sentence** for the judges

2025 GLOBAL AWARD · ART & TECHNOLOGY

A TEAM FROM EGYPT WON.

Zumorroda-X built mini-games where players farm through heat waves and floods using NASA data.

10 Global Winners out of 11,500+ projects from 167 countries and territories. The next one can be from this room.

HOW PROJECTS WIN

JUDGES SCORE FIVE THINGS

INFLUENCE

How big is the problem, and how much does the project help?

AI: a real user, and a decision your model changes

CREATIVITY

Is the approach new, or a small improvement?

AI: one twist: a local angle, a smart feature, an honest finding

VALIDITY

Is it scientifically sound? Would it work for real?

AI: a real metric on data the model never saw

RELEVANCE

Does it answer the challenge? Is it complete and usable?

AI: the model's output shows up in the product

PRESENTATION

Did the team communicate it clearly?

AI: a live demo and one honest sentence

Source: Space Apps Judging & Awards Guide (2024 edition, the latest published; the 2026 guide comes out Nov 13)

HOW PROJECTS WIN

10 GLOBAL AWARDS: AIM AT ONE

Best Use of Science
most valid use of science and the scientific method

Best Use of Data
makes space data accessible, or uses it in a unique way

Best Use of Technology
the most innovative use of technology

Galactic Impact
most potential to improve life on Earth or in the universe

Best Mission Concept
the most plausible concept and design

Most Inspirational
the project that captures our hearts

Best Use of Storytelling
communicates open data through storytelling

Global Connection
connects people around the world through technology

Art & Technology
combines technical and creative skills

Local Impact
greatest potential for local impact

An AI project fits most naturally under the four outlined: Science, Data, Technology, Local Impact.

QUICK VOTE

WHICH TEAM DO YOU TRUST?

TEAM A

“Our revolutionary AI-powered platform understands forests. It predicts tree cover with 86% accuracy.”

TEAM B

“On forest areas it never saw, our model is right 83.5% of the time; guessing gets 7%. It confuses Spruce/Fir with Lodgepole Pine.”

Judges spot “AI-washing” in seconds. Team B gives a number, a baseline and a limit, and every part of it can be checked.

2025 GLOBAL WINNERS THAT USED AI

ONE USER, ONE QUESTION, NASA DATA

BEST USE OF SCIENCE

SpaceGenes+ · Germany

A dashboard showing how radiation and microgravity together affect astronaut health at the molecular level.

BEST USE OF TECHNOLOGY

Twisters · Mexico

SkySense: NASA Earth data plus AI analysis for local weather risk and the safest time for an activity.

BEST USE OF DATA

Resonant Exoplanets · USA

An AI system that analyses telescope data, including JWST spectra, for exoplanets and possible biosignatures.

In each one, the AI is **one part** of a product built around a clear user need.

Source: NASA, “NASA Announces 2025 International Space Apps Challenge Global Winners”, Dec 18, 2025

FROM IDEA TO TASK

START FROM THE PROBLEM, NOT THE MODEL

1 Who is the user?

→

2 What decision do they make?

→

3 What NASA data exists for it?

→

4 What would a model predict?

THE AI TEST

If you removed the model, would anything the user sees change? If not, the “AI” is decoration: cut it, or move it to where it changes the output.

FROM IDEA TO TASK

NAME YOUR TASK TYPE

Task type	Answers	Example	Measure with
Classification	Which category?	tree cover of a 30 m cell; quake or noise	accuracy, precision, recall
Regression	How much?	surface temperature of a block	MAE, RMSE (real units)
Time series	What comes next?	next month's value of a measurement	MAE vs. "same month last year"
Retrieval, generation	What does the evidence say?	answers from NASA papers, with sources	share of test questions answered right

If you can't fill one row for your idea, you don't have an ML task yet.

FROM IDEA TO TASK

BEFORE AND AFTER SCOPING

BEFORE

"We'll use AI to map the world's forests."

AFTER

Predict which of 7 tree covers grows on a 30 m × 30 m cell

From 12 map measurements: elevation, slope, distances, soil

Task classification, 7 classes, labelled by forest surveys

Measure accuracy on areas of the forest it never saw

48 hours? yes: a 15,120-row table, trains in seconds

FROM IDEA TO TASK

FIVE CHECKS BEFORE YOU COMMIT

- **Labels exist**, or you can make a few hundred in an hour
- **The data downloads today**, not after an account approval
- **A baseline trains in minutes** on free Colab
- **One metric** you can compute on data the model never saw
- **The result shows up in your demo**: it passes the AI test

CASE STUDIES

TWO WINNERS THAT SCOPED SMALL

2024 · BEST USE OF TECHNOLOGY

42 QuakeHeroes · Brazil

Signal-processing features plus a small deep network to locate quakes in noisy seismic recordings.

Copy this: one bounded target in noisy data, sized for 48 hours, accuracy described plainly.

2024 · GLOBAL CONNECTION

Asteroid Destroyer · Canada

Data processing plus ML to classify exoplanets by size, temperature and distance.

Copy this: a simple classification from a public dataset, wired straight into a tool people use.

DECIDE HOW TO MEASURE

METRIC, SPLIT, BASELINE: BEFORE ANY CODE

METRIC

Match it to the cost of mistakes

- Balanced classes: accuracy
- A rare class matters (a quake, a flood): precision and recall
- Numbers: MAE or RMSE, in real units

SPLIT

Match it to how the model will be used

Most ML: test rows picked at random (orange)



Related rows (same place, star, person)? Hold out whole groups.

Time series: train past, test future



Never shuffle. Features may only use data from before the moment you predict.

BASELINE

The number your model must beat

- Classes: always guess the most common one
- Numbers: always predict the average
- Time series: same month last year

DECIDE HOW TO MEASURE

PICK THE SPLIT THAT MATCHES REAL USE

Your rows come from…	Split by	Because
one map or image	area (blocks of neighbouring cells)	neighbouring cells are near-copies
a time series	time: train past, test future	the future isn't known when you predict
repeated objects: stars, stations, people	group (whole object)	the same object would sit on both sides
truly independent rows	random	only then is random honest

A random split on rows that aren't independent is **leakage**: the model sees near-copies of the answers.

CASE STUDY

STORM PROPHET: A FORECAST, FRAMED RIGHT

2023 · BEST USE OF DATA · UKRAINE

Develop the Oracle of DSCOVR

An LSTM model trained on solar-wind data to predict geomagnetic storm levels.

COPY THIS

- Frame it as a time series
- Train on the past, test on the future
- Compare with a simple baseline
- Show confidence and limits, not just one number

LIVE NOTEBOOKS · 38 MIN

FOREST COVER, END TO END

[El-Moghazy2/space-apps-forest-cover-colab: Forest Cover Type: a standalone Colab notebook for the AI & ML workshop at NASA Space Apps Cairo 2026]

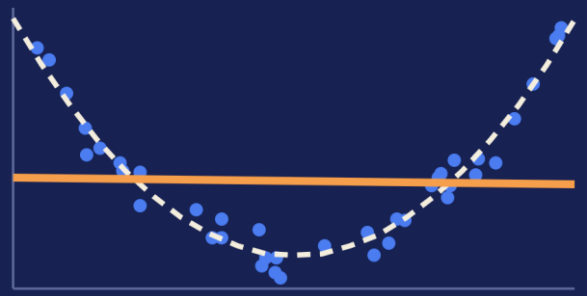
Predict the tree cover of a 30 m patch of forest from 12 map measurements

Notebook · step	What it shows
Understand · 1–3	What a row is; 44 columns hide 2 categories
Understand · 4	Train is balanced, the real forest isn't
Understand · 5–8	Elevation dominates; combinations beat raw columns
Understand · 9–10	Rows are map neighbours, not independent
Model · 1	Test on unseen forest areas, not random rows
Model · 2–3	Model families; correcting the class balance
Model · 4–5	Each feature idea, before and after
Model · 6–8	Optuna, number of trees, the whole journey

MODEL LIMITATIONS

EVERY MODEL HAS A BLIND SPOT

Linear: lines only

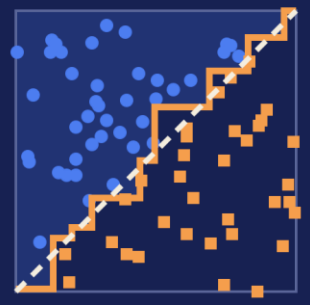


Dots follow a U. Solid: linear regression. Dashed: with an x^2 feature.

Fix: add curved features (x^2, distance from ideal), or use a tree

R^2 on new data: -0.02 → 0.92

Trees: diagonal = stairs

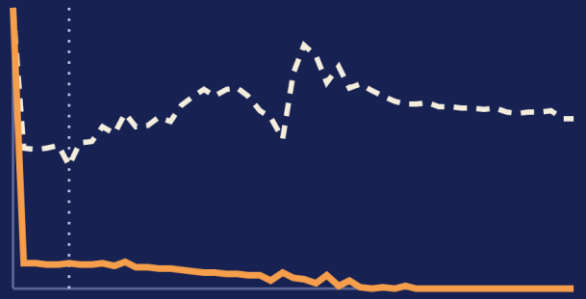


Circles above a 45° line (dashed), squares below. Solid: the tree's splits.

Fix: give it the combination (y - x) as a column: one split

17 leaves, 95% → 2 leaves, 99.8%

Neural nets: memorise



12 points, 3 layers. Solid: training error. Dashed: new data. Dotted: stop here.

Fix: more data, augmentation, early stopping, or a pretrained model

Train error 0.0002; new data 0.11 → 0.15

TREE LIMITATIONS · REAL LABELS

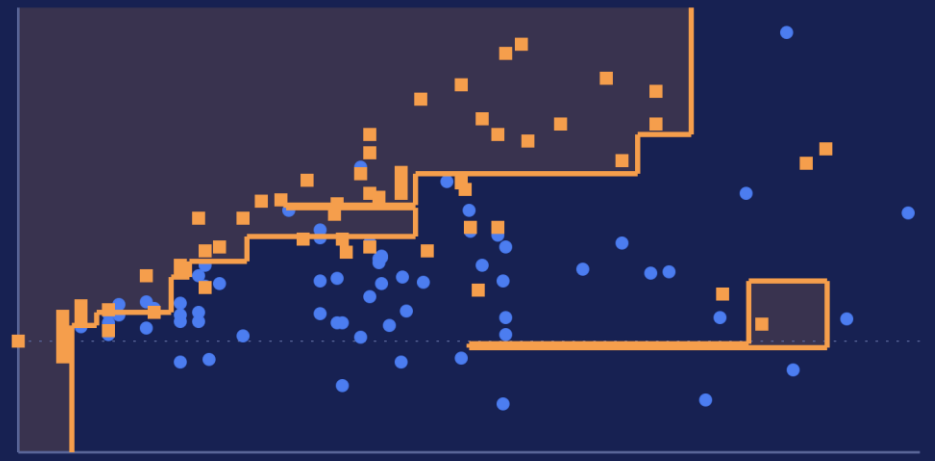
TWO COLUMNS, ONE AXIS, ONE CUT

Real cells, real labels, the same cells in both panels. Circles: Lodgepole Pine. Squares: Cottonwood/Willow. Accuracy on areas of the forest the model never saw.

Before: horizontal + vertical

85.2% · size chosen by cross-validation

31 splits

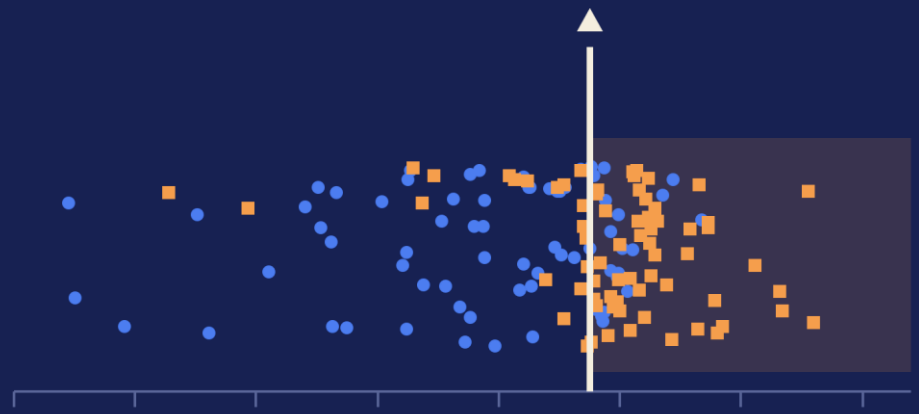


h across (0–600 m), v up (-80–240 m) · tiny steps hidden

After: one column, v − 0.4 × h

85.3% · cut at -12 m

1 split

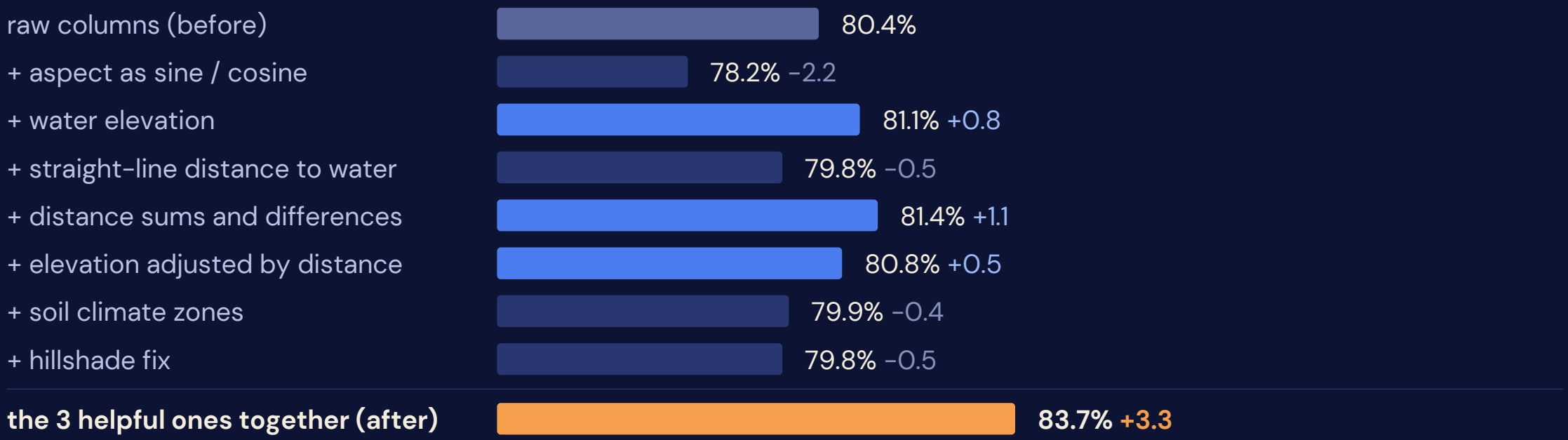


v − 0.4 h across (−250–120 m) · spread only avoids overlap

Trained on only 30 cells: 71% vs **84%**. Straight-line distance $\sqrt{(h^2 + v^2)}$ didn't help: one split on it = one split on h (81.1%).

RECAP FROM THE NOTEBOOK · STEP 4

ONE FEATURE IDEA AT A TIME



TREE LIMITATIONS · WHAT WE MEASURED

FIVE LIMITS OF TREE MODELS

Limit	On the forest data	Fix
One column per question: diagonals become stairs	water elevation alone +0.8; three combinations together +3.3	add differences, sums, ratios
They learn the training class balance	balanced train, real forest 50% Lodgepole: expected 69.7% → 77.5% once corrected	re-weight to the real class shares
Textbook encodings can hurt	aspect as sine/cosine: -2.2 points for extra trees	measure, don't assume
Bad settings cost a lot	worst of 30 Optuna trials 77.2% vs default 83.7%	start from defaults, tune last
Can't predict beyond what they've seen	a trend forecast stays under the training maximum	remove the trend first

RECAP FROM THE NOTEBOOK · STEP 1

WHICH NUMBER DO YOU REPORT?

86.4%

Cross-validation on shuffled rows

85.6%

A test set of random rows

80.4%

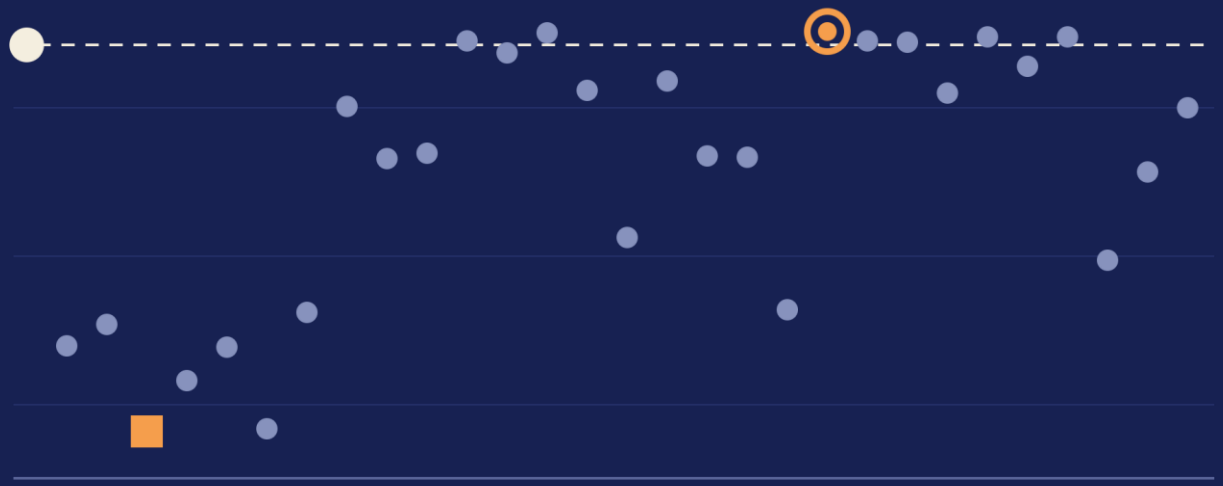
Forest areas it never saw.
The honest number.

Same model, same data: only the test changed. Random rows let the model peek at the neighbours of every test cell.

RECAP FROM THE NOTEBOOK · STEP 6 · TUNING WITH OPTUNA

TUNING: PROTECTION, NOT A BOOST

30 settings Optuna tried for extra trees



Each dot is one setting, scored by cross-validation on the training data (up: 77% → 83.5%; lines at 78, 80, 82%). **Cream dot and dashed line: default.** Ring: best. Square: worst.

Setting	Cross-val.	Local test
Default	82.8%	83.7%
Best by Optuna	83.0%	83.5%
Worst	77.6%	77.2%

Bad settings cost 6.5 points

The worst setting vs the default, on unseen forest areas.

The best is just noise

+0.2 in cross-validation, −0.2 on the test. Keep the defaults; feature engineering gave +3.3.

GENAI AND FOUNDATION MODELS

A FOUNDATION MODEL IS STILL A MODEL.

Same split. Same metric. It has to beat your simple baseline.

GENAI AND FOUNDATION MODELS

THREE YOU MIGHT USE, AND WHAT EACH MUST BEAT

TIME SERIES

Chronos (Amazon)

Pretrained on many time series, so it forecasts without training on yours (zero-shot).

Must beat: “same month last year”, on a time split

SATELLITE IMAGERY

Prithvi (NASA, IBM)

A geospatial foundation model: strong image features from fewer labels.

Must beat: your simple baseline, on the same area split

TEXT AND DOCUMENTS

LLMs + RAG

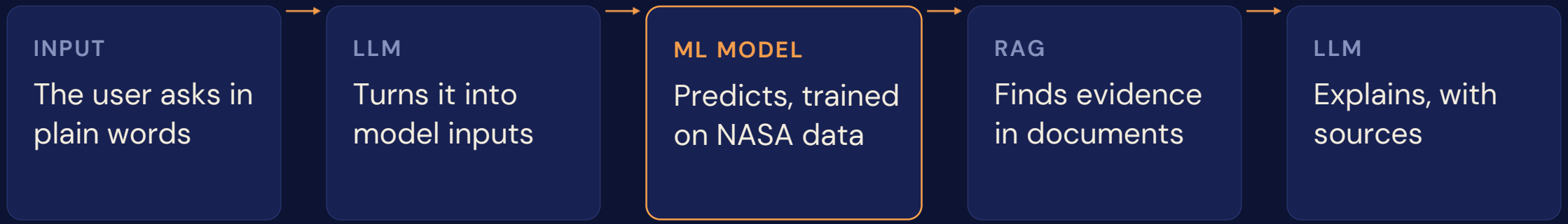
Reads the user’s question, answers from documents with citations, explains a prediction.

Must pass: about 10 test questions with known answers

Same rule as every model today: beat the simple baseline, on an honest split.

GENAI AND FOUNDATION MODELS

A STRONG DESIGN: MODEL + LLM



Numbers come from the model. Facts come from the documents. Words come from the LLM.

GENAI AND FOUNDATION MODELS

AI TOOLS AT THE HACKATHON: THE RULES

ALLOWED

Coding assistants, chatbots, image tools, foundation models

REQUIRED

Describe how you used AI tools in your project submission

NOT ALLOWED

Starting work on your challenge before the event (Nov 14)

NEVER

Report a number your own code didn't compute

Source: NASA Space Apps 2026 Participant FAQs (the FAQ calls these tools "Super Intelligence" tools)

TRAIN LESS · TRANSFER LEARNING

DON'T TRAIN WHAT SOMEONE ALREADY TRAINED

TRANSFER LEARNING: REUSE A PRETRAINED MODEL'S KNOWLEDGE

1 · ZERO-SHOT

Use a pretrained model as it is: Chronos forecasts your series, an LLM labels your text.

Minutes
no training labels

2 · FROZEN MODEL + CLASSIFIER

Run the data through the pretrained model **once**, save its outputs (embeddings), train trees or logistic regression on them.

Minutes, then seconds per retrain
a few hundred labels

3 · FINE-TUNE THE LAST LAYERS

Replace the last layer with one for your classes; retrain it (and a few layers before it) on your data.

Under an hour on a free GPU
hundreds to thousands

4 · FINE-TUNE EVERYTHING

Update every weight of the pretrained model.

Hours: overnight at most
thousands of labels

5 · FROM SCRATCH

Random weights, learn everything from your data.

Days: never in 48 hours
very large datasets

Climb only if your rung can't beat the baseline. **Rung 2 first:** the slow part runs once; every experiment after it takes seconds.

TRAIN LESS · BEST PRACTICES

START FROM A MODEL TRAINED ON DATA LIKE YOURS

Your data	Start from	How (rung of the ladder)
Satellite images	Prithvi-EO-2.0 (NASA, IBM)	transfer learning: frozen + classifier (2), or fine-tune with TerraTorch (3)
Weather and climate	Prithvi WxC (NASA, IBM)	large: fine-tune the last layers on a GPU (3)
The Sun	Surya (NASA, IBM)	fine-tune for flares or solar wind (3); trained on 9 years of SDO images
Time series	Chronos-Bolt (Amazon), TimesFM (Google)	zero-shot forecasts, no training (1)
Text, papers	Sentence embeddings (e.g. MiniLM)	frozen, on a CPU: the search step of RAG (2)
Ordinary photos	ImageNet models, e.g. ResNet, EfficientNet (timm, torchvision)	transfer learning: frozen + classifier (2), then fine-tune the last layers (3)
Tables	Nothing	no transfer needed: extra trees train in seconds

All on Hugging Face. **On day 1, check the size, the licence and that it runs on free Colab**, before you build on it.

TRAIN LESS · BEST PRACTICES

FOUR RECIPES THAT FIT IN A HACKATHON

SATELLITE IMAGES

Burned or not, per HLS image tile

Transfer learning: frozen Prithvi-EO-2.0 (tiny) turns each tile into a vector, once; logistic regression learns from the vectors. Not enough? Fine-tune its last layers.

Rung 2, then 3 · one pass on a free GPU, then seconds

Must beat: a burn-index threshold, on held-out areas

TIME SERIES

Next 12 months of NASA's global temperature

Give Chronos-Bolt the monthly GISTEMP history; it returns a forecast with a range. Nothing to train.

Rung 1 · seconds, on a CPU

Must beat: "same month last year", on the last 2 years

TEXT

Answer questions from mission documents

Cut the documents into passages, embed them once, retrieve the closest ones, let an LLM answer with citations.

Rungs 1 + 2 · minutes, on a CPU

Must pass: 10 questions with known answers

TABLES

Tree cover per 30 m cell (today)

No pretrained model needed: extra trees on the columns, plus a few engineered features.

From scratch, but seconds on a CPU

Beat: guessing (14%) with 83.7%, on unseen areas

TRAIN LESS · BEST PRACTICES

KEEP EVERY TRAINING RUN SHORT

Prototype on a sample

Use 5–10% of the data until the whole pipeline runs end to end. Scale up once, at the end.

Compute once, save it

Save embeddings and features to disk (.npy, .parquet). A restart then costs seconds, not an hour.

Fast baseline first

Trees train in seconds. A slow deep model has to beat them to stay in the project.

Shrink the inputs

Lower resolution, fewer bands, shorter windows, the smallest model version. Mixed precision on a GPU.

Stop early, save often

Stop when the validation score stops improving. Save a checkpoint every epoch: a Colab disconnect costs minutes.

Time-box the tuning

Tune last, with a fixed budget, e.g. Optuna with a n-minute timeout.

Free GPUs: Google Colab and Kaggle Notebooks (weekly quota). **Any run longer than 15 minutes: start it before a break or before bed, never while the team waits.**

YOUR TURN · 4 MINUTES

FILL IN YOUR SCOPE CARD

User and decision	who, and what do they decide with it?
Predict … from …	one target, from which NASA data?
Task type	classification, regression, time series, retrieval
Metric	and why it matches the cost of mistakes
Split	random, by time, by area or by group, and why
Baseline to beat	the simplest guess, and its score

YOUR TURN · THEN 2–3 READ ALOUD

THE HONEST CLAIM, IN ONE SENTENCE

FORMULA

Our [model] gets [metric] on [data it never saw] (baseline: [score]). It struggles with [limit], and hasn't been tested on [scope].

EXAMPLE FROM THE NOTEBOOK

"On areas of the forest it never saw, our model labels 83.5% of 30 m cells correctly (guessing gets 7%). It confuses Spruce/Fir with Lodgepole Pine, and it was trained on one Colorado forest, so it is untested anywhere else."

THE HACKATHON PLAN

SOMETHING WORKING BEFORE YOU SLEEP

DAY 1 · MORNING

Problem, user, target award. Scope card and the five checks. Download the data.

DAY 1 · AFTERNOON

Explore. Fix the metric, split and baseline.

DAY 1 · EVENING

One full pipeline: data → model → simplest demo.

NIGHT

Sleep. Tired teams leak data and break demos. A long training run can go overnight.

DAY 2 · MORNING

Improve: features, error analysis. GenAI only if it changes the output. **Freeze the model by lunch.**

DAY 2 · AFTERNOON

Honest claim, limits, AI-use description. Demo video, slides, rehearse. **Submit early.**

Everything after day 1 is a bonus. A simple model that works beats a clever one that doesn't.

IF YOU REMEMBER ONLY FIVE THINGS

TAKE THESE HOME

1. Scope one small ML task before writing any code.
2. Evaluate on data the model hasn’t seen, split the way it will be used.
3. Report one real metric: never a number you didn’t calculate.
4. Say what your model can’t do as clearly as what it can.
5. A small, honest model beats a vague “AI-powered” claim.

30 MINUTES

QUESTIONS

AbdElRhman ElMoghazy

https://www.linkedin.com/in/abdelrhman-elmoghazy/

RESOURCES

Today's notebooks: understand + model

2026 challenges · spaceappschallenge.org

Participant FAQ: AI-use rules

Judging & Awards Guide: out Nov 13